Forecasting of the ARMA(p, q) Process

$$
\begin{aligned}
&\mathbb{E}\left[\left(X_{n+k} - \hat{X}_{n+k}(n)\right)^2\right] \\
&= \mathbb{E}\left[\left(\sum_{j=0}^{k-1} b_j Z_{n+k-j} + \sum_{j=0}^{\infty}\left(b_{j+k} - c_j(k)\right) Z_{n-j}\right)^2\right] \\
&= \mathbb{E}\left[\left(\sum_{j=0}^{k-1} b_j Z_{n+k-j}\right)^2\right] \\
&\quad + \underbrace{\cancel{2\,\mathbb{E}\left[\left(\sum_{j=0}^{k-1} b_j Z_{n+k-j}\right)\left(\sum_{j=0}^{\infty}\left(b_{j+k} - c_j(k)\right) Z_{n-j}\right)\right]}}_{0 \text{ (no } Z_t^2 \text{ terms)}} \\
&\quad + \mathbb{E}\left[\left(\sum_{j=0}^{\infty}\left(b_{j+k} - c_j(k)\right) Z_{n-j}\right)^2\right] \\
&= \mathbb{E}\left[\left(\sum_{j=0}^{k-1} b_j Z_{n+k-j}\right)^2\right] \\
&\quad + \underbrace{\mathbb{E}\left[\left(\sum_{j=0}^{\infty}\left(b_{j+k} - c_j(k)\right) Z_{n-j}\right)^2\right]}_{\text{This is zero if } b_{j+k} - c_j(k) = 0}
\end{aligned}
$$

$\therefore$ To minimise the mean square prediction error, we choose $c_j(k) = b_{j+k}$.